Rejuvenation readouts cannot distinguish younger from less old

A one-dimensional age score falls for any move away from the old state. Distance to a young target of the same type tells approach from departure.

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SUMMARY

Treatments said to rejuvenate cells are usually judged by a single score that falls when cells look younger. But such a score shows only which way cells moved, not where they ended up: it also falls when cells simply stop looking old, for example by drifting toward a different cell type, without becoming any more like young cells. We ask a more direct question: after treatment, are the cells closer to young cells of the same type than they were before? In skin fibroblasts from a 96-year-old donor that were partially reprogrammed (Lu et al.), the published aging score fell well below the level of a 22-year-old donor's untreated cells, which looks like strong rejuvenation. Yet the same cells ended about twice as far from the young donor's cells as they started, even when judged only by the genes that make up that score. The same was true with other published aging genes, with all genes, with hundreds of randomly chosen gene sets, and with an age score we built that follows donor age across several independent cohorts. Our measure does register cells getting closer when young cells are mixed into old ones, so it can see real rejuvenation when it happens, and the result is not the two donors' cells simply becoming more alike. A falling aging score therefore cannot tell rejuvenation apart from cells moving away from their old state in some other direction. We provide a frozen age score, young reference points and code for measuring distance to young cells instead.

INTRODUCTION

Partial reprogramming and other candidate rejuvenation interventions are evaluated by asking whether a cell's molecular profile has moved opposite to the direction of aging. Epigenetic clocks, curated aging signatures and correlation-based reversal scores share this form: a single number that falls when the cell is judged younger.

This form has a structural weakness: it records the direction a cell moves and not where it ends up. In the high-dimensional expression space of more than twenty thousand genes, almost every direction leads away from the old state, and only a narrow cone of them leads toward young cells of the same type. A score anchored on not-old falls across that whole space. In that space a displacement ends closer to the young target only when its length is less than twice the starting distance to the target times the cosine of the angle between the displacement and the target direction. A directional score carries the angle and discards the length, so it cannot evaluate that condition even in principle. Three outcomes therefore become hard to tell apart:

- Approach to young. The cell becomes more similar to a young cell of the same type and ends closer to it than it began. This is the intended outcome.
- Departure in the wrong direction. The cell becomes less similar to an old cell by becoming something else. Loss of the differentiated state is one such move, and the one with the clearest cost: in the limit it is dedifferentiation and loss of tissue function. It is not the only one. A shift in proliferative state, a stress response, or a change of subtype leaves the cell no closer to young while scoring just as well.
- Departure past the target. The cell moves in the right direction and does not stop, ending beyond the young state rather than at it. This scores highest of all on a directional readout, and it is the failure mode of complete reprogramming.

The distinction is concrete for reprogramming. Complete OSKM reprogramming proceeds through a mesenchymal-to-epithelial transition and ends in pluripotency; the whole trajectory moves away from the aged fibroblast state, and a score anchored only on not-old will register the whole of it as rejuvenation.

We therefore replace the direction with a target. For a given cell type there is a region of expression space that the intervention should approach: this cell type, young. Movement toward it is progress; movement past it, or orthogonal to it, is not. This turns identity from a competing axis into a stopping condition, and turns age from a direction into a distance.

Operationalizing this requires three things: a young target of the same cell type, measured on the same platform where possible; a statistic that separates closer-to-young from farther-from-old; and controls that detect the confounds which make the two look alike. Two controls do most of the work here. A dose-response positive control, in which known amounts of young cells are mixed into old ones, establishes that the statistic can see an approach at all and sets the noise floor as the value observed at zero dose. A reverse test, in which the roles of the young and aged samples are swapped, distinguishes rejuvenation from mutual convergence of two samples. We begin with the published readout whose behavior prompted the question, show that the behavior does not depend on which genes are read, then build an age ruler that outperforms that readout on donor age and show that the ruler is not exempt either.

RESULTS

A published aging score falls while the cells move away from young cells

Lu et al. use a mesenchymal-drift (MD) signature, the Hallmark epithelial-mesenchymal transition gene set augmented with SNAI1, ZEB1, ZEB2, TWIST1 and TWIST2 (205 genes), as their readout of aging, and report that it falls when aged fibroblasts are partially reprogrammed. Their dataset GSE297234 is an OSKM reprogramming time course in dermal fibroblasts from a 96-year-

old donor (GM00731) and a 22-year-old donor (GM23815), sampled at days 0, 3, 7 and 10. It is unusual among rejuvenation datasets in containing young cells of the same type, on the same platform, in the same experiment, which is what makes the question in this paper answerable. Cell states were labeled with the published state signatures (Methods). The population tested is the aged donor's partially reprogrammed cells (5,832 cells). They are overwhelmingly day-3 cells (5,220 of them), assayed while OSKM is still being expressed; this is the population the source paper treats as rejuvenated. Donors are never pooled, and the 22-year-old donor is a contrast, not a replicate.

We scored MD per cell with a reimplementation of the published algorithm (Methods), scoring both donors against the same control genes so that their scores are comparable. The score behaves as reported. In the aged donor it falls from 0.515 (95% cell-resampling interval 0.513 to 0.517; these intervals describe within-sample precision, not donor-level uncertainty) in day-0 fibroblasts to 0.158 (0.153 to 0.162) in partially reprogrammed cells, and continues downward through early pluripotency (0.147) to pluripotency (0.022). The relevant comparison is the difference between the two donors' day-0 fibroblasts, which is what 74 years of donor age look like on this scale: 0.046 (0.043 to 0.049). The drop on partial reprogramming is 7.7 times that difference (95% CI 7.3 to 8.2). Read as an age readout, the 96-year-old's partially reprogrammed cells are far younger than the 22-year-old's actual cells (Figure 1A).

We then asked where those cells are. Each population's counts were summed and normalized, and each gene was standardized against its mean and spread across cultured dermal fibroblasts from 652 adult GTEx donors, which serve throughout as a fixed reference and define a panel of 23,485 expressed genes (Methods). Restricted to the 205 MD genes, and with the young donor's untreated day-0 cells as the target, the aged donor's day-0 cells start 14.6 units away. After partial reprogramming they are 31.7 away: 2.2 times farther than they began. They cover 79% of the line from the aged donor's cells to the young donor's (progress 0.790, 95% CI 0.780 to 0.811), which a projection-based readout would score as substantial rejuvenation, but they also move 2.16 times that line's length off to the side. Covering most of the line is therefore compatible with ending much farther away (Figure 1B). The cells also travel a long way: 2.3 times the line between the two donors. A move longer than twice that line cannot end closer to the target whichever way it points, so these cells overshoot on length alone. Complete reprogramming goes further in the same way (pluripotent cells: progress 1.33, final distance 3.9 times the starting distance).

Two checks decide whether this is a real result or an insensitive measure. First, mixing known fractions of the young donor's day-0 cells into the aged donor's day-0 cells, a sample that genuinely moves toward young, gives steadily increasing progress (0.010, 0.096, 0.207 and 0.388 at 0%, 10%, 25% and 50%) and steadily decreasing distance to the young target (−0.007, −1.195, −2.676 and −5.143), with intervals that contain their point estimates. The measure sees a real approach of this size when one occurs. Second, with the roles of the donors reversed, the young donor's partially reprogrammed cells do not move toward the aged donor (progress −0.229, 95% CI −0.251 to −0.224) and end 2.7 times farther from it, so the forward result is not the two donors' cells becoming more alike.

The reason the score still falls is visible in the same genes. Approximating MD by the direction in which every MD gene decreases, the cells' displacement aligns with that direction at cosine 0.53, whereas the difference between the two donors' day-0 fibroblasts aligns with it at only 0.20. The score rewards a pattern that the donors' age difference resembles only weakly. MD genes are largely mesenchymal identity genes, so a falling MD score is in the first instance a statement about how fibroblast-like a cell remains, which is consistent with what Lu et al. describe.

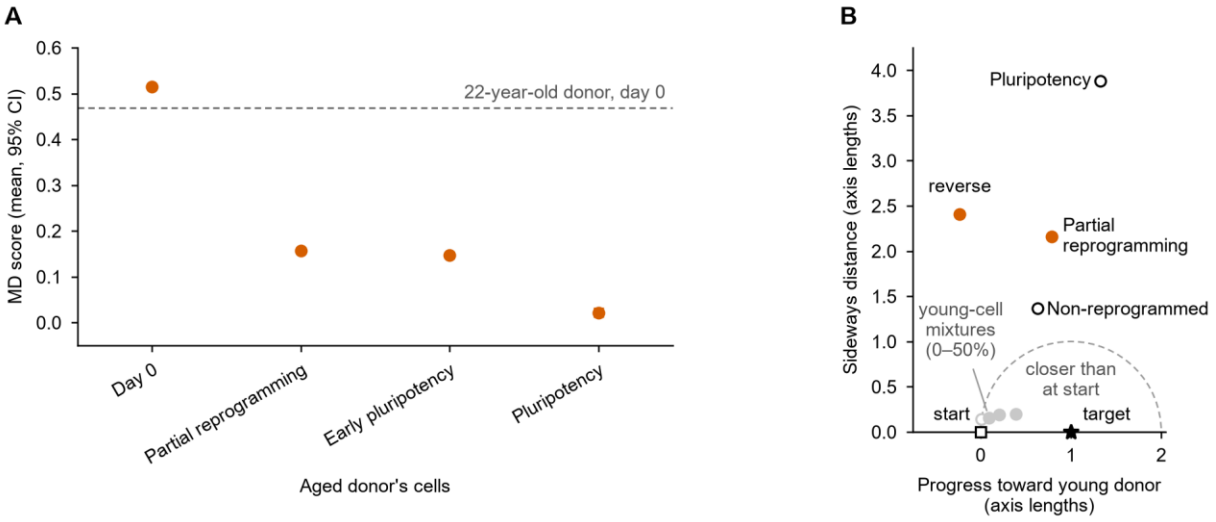


Figure 1. A published aging score falls while the cells move away from young cells. (A) MD score in the aged donor's cells by state, mean and 95% bootstrap interval over cells, with both donors scored against the same control genes. The dashed line is the 22-year-old donor's day-0 fibroblasts: the same score applied to cells that really are young. (B) The same cells in the 205 MD genes, drawn in the plane spanned by the line from the aged donor's day-0 cells (square, at 0) to the young donor's day-0 cells (star, at 1) and the direction perpendicular to it, in units of that line's length. Points inside the dashed circle are closer to the young target than the starting point was. Gray points are young-cell mixtures (0 to 50% young cells; open, 0%); the reverse test is drawn in its own frame with the donors swapped. Donors are never pooled.

The result does not depend on which genes are read

One reading of Figure 1 is that MD is simply the wrong list of genes, and that a list chosen for age rather than for mesenchymal identity would behave differently. We repeated the test in other gene spaces, rebuilding the target, the mixtures and the reverse test inside each one: the age-up and age-down lists that Lu et al. report alongside MD, derived from Fleischer et al. (3,089 of their 3,538 unique gene symbols are in the reference panel; the two published lists hold 1,533 and 2,007 entries, with one duplicated symbol each); all 23,485 genes of the panel; and, to ask whether any list at all would show an approach, 200 random gene sets of 205 genes and 200 of 3,089 genes, each matched to the real sets on expression level and on whether the genes are detected in these cells.

The answer is the same everywhere (Figure 2). In the age genes the partially reprogrammed cells cover 82% of the line toward the young donor and end 1.9 times farther from the young target than they started; in all genes, 67% and 2.0 times. Of the 400 random gene sets, none ends closer to the young target, in either direction of the test; the closest any comes is 1.81 times its starting distance. In all 400 sets the cells travel at least 1.95 times the length of the donor line, and in none of them is the move short enough to end closer given the direction it takes, so the overshoot does not depend on which genes are read. All 400 nonetheless register the 50% young-cell mixture as moving closer, so a gene set of this size is not too small to detect an approach.

The age lists are not arbitrary, which strengthens the point rather than weakening it. Their progress along the line toward the young donor, 0.822, is higher than that of all 200 random sets of the same size, so these genes do capture more of the difference between the two donors than random genes do, and the cells still end farther away.

Nor does the result depend on the other choices the test makes. The young target is a single donor, so we also defined young as the GTEx reference donors aged 20 to 39 and old as those aged 60 to 79. In all genes and in the MD genes, the partially reprogrammed cells end farther from the young group and farther still from the old one: they recede from both. In the age genes they end slightly closer to both, more so to the old group, which is movement toward fibroblasts in general rather than toward young ones (Supplementary Table S1, Supplementary Figure S1). This comparison crosses the gap between single-cell and bulk sequencing, so it shows departure clearly but could not show a small approach (Supplementary Note 1). Relabeling the cells, by clustering both donors together as Lu et al. did or by using their published per-cell labels, left every conclusion unchanged (Methods).

The result does not depend on the coordinate system

A second reading is that the overshoot comes from the coordinates rather than the cells. Every distance so far is computed in log2 CPM, z-scored with the GTEx training mean and standard deviation, and both steps are choices. The log compresses a million-fold range of abundance into a few units. The z-score divides each gene by its spread between donors, so a gene that barely varies between people can contribute a large coordinate from a small change. We repeated the same-platform test, pre-registered as a secondary analysis, in two other coordinate systems: counts per million with no log and no z-score, in which abundant genes dominate, and log2 CPM with no z-score, in which distances are fold changes. The panel, the TMM factors, the day-0 split, the mixtures and the random draws were held fixed, and nothing was refit. A coordinate system counted only if the young-cell mixture control detected known young fractions in it, and all three did.

The answer is the same in each (Supplementary Table S2). The partially reprogrammed cells end 2.05 times farther from the young donor than they started in counts per million, 1.88 times in log2 CPM and 2.00 times in the main analysis. In every system the asymmetry between the forward and reverse tests is positive, so

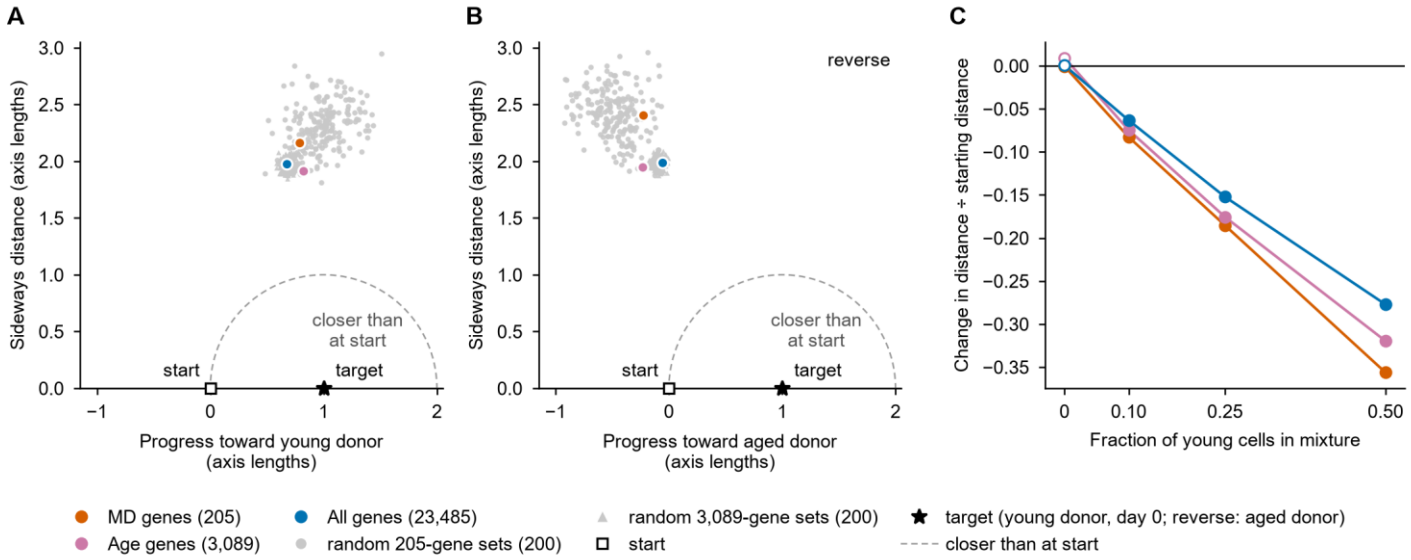


Figure 2. The result does not depend on which genes are read. (A) The test drawn as in Figure 1B for the MD genes (205), the published age-up and age-down genes (3,089) and all genes of the reference panel (23,485), with 200 random gene sets of each of the two smaller sizes matched on expression level and detection. Points inside the dashed circle are closer to the young target than the starting point was. (B) The reverse test, with the donors swapped. (C) Young-cell mixtures in the three gene spaces: change in distance to the young target divided by the starting distance, against the fraction of young cells mixed in; open symbols are the zero-mixture noise floor. No gene set ends inside the circle in A or B, and every gene set registers the mixtures as moving closer in C.

the movement is not mutual convergence in any of them. The coordinates change how far the cells appear to travel along the donor line (progress 2.05, 0.58 and 0.67), but in none of them is the move short enough to end closer to the young donor.

Without the log the agreement is narrower than it looks. Ten highly expressed transcripts, among them MALAT1, VIM and FTL, carry 77% of the squared final distance, against 3.6% in log2 CPM and 1.8% in the main analysis. In that system the result is a statement about a handful of genes, and it agrees in sign with the others without carrying the same weight (Supplementary Note 3).

A frozen fibroblast age ruler orders donors by age

One objection remains: MD may simply be a poor measure of age, and a score that genuinely tracked donor age might not be misled. To test this we built an age score from public data. We trained a ridge-regression ruler on donor chronological age from all 23,485 genes of the reference panel, in the same 652 GTEx fibroblast donors, and froze its weights, training mean and standard deviation before any evaluation. The frozen ruler was then applied unchanged to held-out donors and to two external cohorts that differ in laboratory, protocol and sequencing platform (Table 1, Figure 3).

Two qualifications apply. First, transfer is an ordering, not a calibrated age. In the external cohorts the frozen score is a unit-length projection with no intercept or scale, so only the ordering can be evaluated, and we never use the ruler as an age estimate. Second,

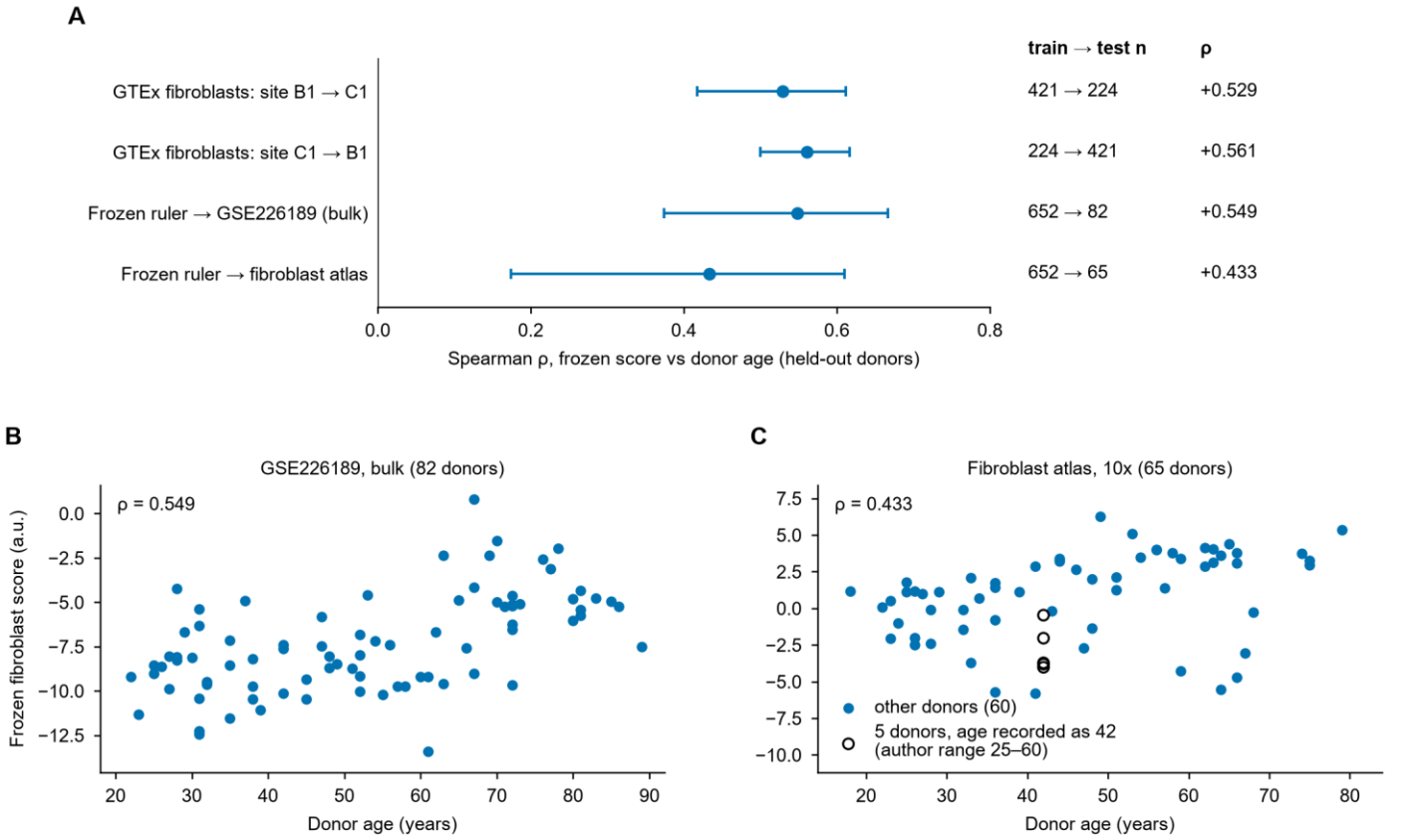


Figure 3. Transfer of the frozen fibroblast ruler. (A) Spearman ρ with donor age for every transfer in Table 1, with 95% percentile bootstrap intervals from 200 resamples of donors. (B) GSE226189 bulk cohort, 82 donors. (C) Dermal fibroblast atlas, 65 donors. Repeated samples from the same person are averaged; the 23 accessions from five donors of one constituent study, whose age is recorded as 42 but reported by the source authors only as 25 to 60 years, are drawn as open circles.

Table 1. Transfer of the frozen fibroblast ruler to held-out donors. ρ is Spearman correlation between the ruler score and donor age. The ruler orders donors; it is not a calibrated age in years. External cohorts are scored with the frozen unit-length projection (the direction of the fitted weights only: no added constant and no conversion into years). GTEx ages are public only as decade bins, so the GTEx rows correlate against bin midpoints. The dermal fibroblast atlas is scored with one value per person (65 donors).

Training set	Held-out cohort	n train → test	Spearman ρ (score vs age)	Permutation null
GTEx fibroblasts, site B1	GTEx site C1	421 → 224	+0.529	p = 0.005
GTEx fibroblasts, site C1	GTEx site B1	224 → 421	+0.561	p = 0.005
GTEx fibroblasts, 652 donors (frozen)	GSE226189, bulk	652 → 82	+0.549	p = 0.005
GTEx fibroblasts, 652 donors (frozen)	Dermal fibroblast atlas, 10x, 65 donors	652 → 65	+0.433	p = 0.005

the permutation nulls shuffle donor ages with the expression matrix held fixed; p = 0.005 is the smallest value attainable with 200 permutations.

One of these transfers is weaker than it looks. The dermal fibroblast atlas integrates published studies; tracing each sequencing identifier to its source shows 65 people, including five donors whose recorded age of 42 is a placeholder for an author-reported range of 25 to 60 years. Scores were averaged within person before correlation (ρ = +0.433, 95% CI +0.174 to +0.610). The bulk cohort GSE226189, with 82 distinct donors, is the cleaner external test.

The ruler tracks donor age better than MD

If MD is to serve as an age readout, it should predict donor age. We compared three frozen instruments, none of them refit here, on two fibroblast cohorts that none of them was fitted on: the ruler, MD, and the difference between the age-up and age-down scores (Table 2, Figure 4).

The evidence is therefore uneven, and we state it that way. In GSE226189 the ruler beats both curated instruments clearly, and neither MD nor the age-up and age-down lists order donors better than chance there (MD permutation p = 0.26). In the fibroblast atlas the ruler still beats the age-up and age-down lists (+0.372, 95% CI +0.094 to +0.595), but its advantage over MD (+0.171, CI −0.118 to +0.412) is not distinguishable from zero. The claim that survives is that the ruler tracks donor age better than MD in one clean cohort and at least as well in a second. MD was scored with the authors' gene list and a reimplementation of their algorithm (Methods).

Gene coverage also differed between instruments. The MD list was almost complete in both cohorts (202 and 204 of 205 genes), but the age-up list was substantially incomplete (986 and 1,040 of 1,533 genes in the atlas and GSE226189), as was the age-down list (1,595 and 1,745 of 2,007), so the ruler's advantage over the age-up and age-down lists may partly reflect missing genes rather than a weaker signature. The comparison with MD is not affected by this.

By the standard an age readout must meet, the ruler is at least as good as MD, and better in the cleaner cohort. What remains is whether that protects it from the failure shown in Figures 1 and 2.

A better age score makes the same mistake

The ruler reads the same 23,485 genes in which Figure 2 already places the reprogrammed cells, so its score and their position can be set side by side. On the ruler, the aged donor's day-0 cells score +3.22 and the young donor's day-0 cells −1.89. The aged donor's partially reprogrammed cells score +0.65, half of the way to the young donor; its early pluripotent cells score −2.62, already younger than the young donor; and its pluripotent cells score −10.40, far younger still (Figure 5A). In the same genes, the partially reprogrammed cells end twice as far from the young donor's cells as they started (Figure 2A), and the pluripotent cells nearly four times as far. The more strongly the ruler says the cells are getting younger, the farther they are from young cells. It is the same failure as MD's, in an instrument built to track donor age.

Part of the reason is a property of the ruler itself. Its score is anti-correlated with distance from its own training distribution: across 22 pseudobulks in the reprogramming dataset (one per donor, cell-

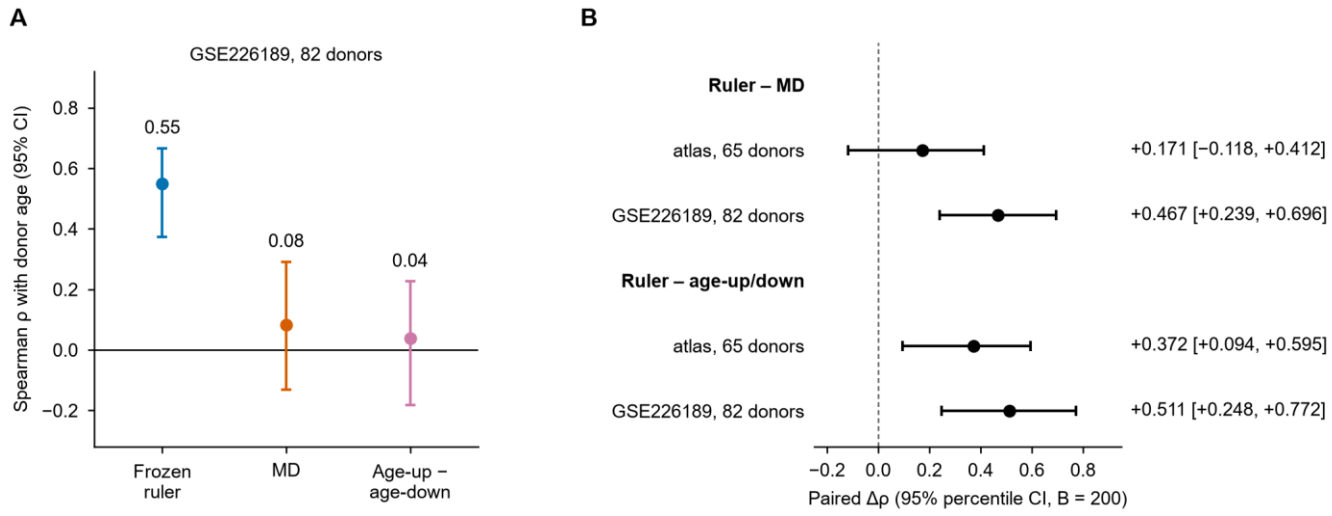


Figure 4. The frozen ruler against two curated instruments. (A) Spearman ρ with donor age for each instrument in GSE226189, with 95% percentile bootstrap intervals. (B) Paired Δρ (ruler minus the other instrument). Against MD, the atlas interval includes zero.

Table 2. Three frozen instruments against donor age in two external fibroblast cohorts. Δρ is paired: both correlations are recomputed on the same resampled donors (B = 200), so the interval is on the difference itself rather than on the overlap of two separate intervals. Atlas scores are one value per person (65 donors).

Cohort	Ruler ρ	MD ρ	Age-up − age-down ρ	Δρ (ruler − MD) [95% CI]	Δρ (ruler − age-up/down) [95% CI]
Fibroblast atlas, 65 donors	0.433	0.26	0.06	+0.171 [−0.118, +0.412]	+0.372 [+0.094, +0.595]
GSE226189, bulk (82 donors)	0.549	0.082	0.037	+0.47 [+0.24, +0.70]	+0.51 [+0.25, +0.77]

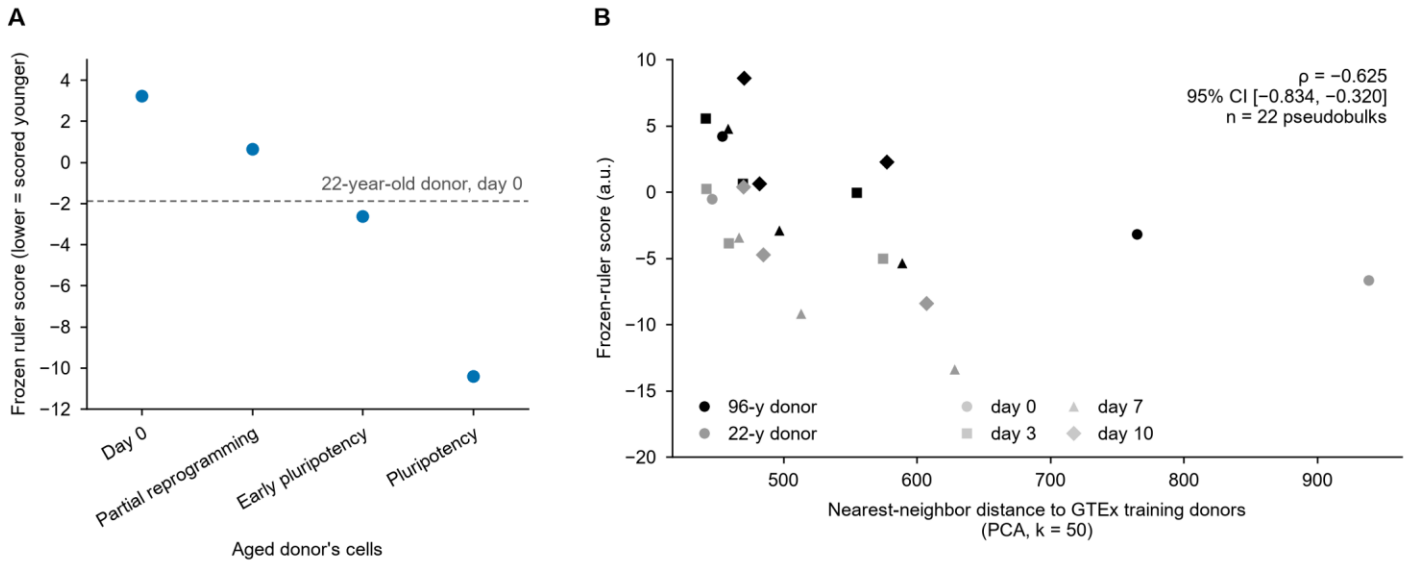


Figure 5. A better age score makes the same mistake. (A) Frozen ruler score of the aged donor's cells by state; lower is scored younger. The dashed line is the 22-year-old donor's day-0 fibroblasts. Each value is one pseudobulk, with each donor's states normalized together. In the same 23,485 genes, the partially reprogrammed cells end twice as far from the young donor's cells as they started (Figure 2A). (B) The ruler scores cells far from its training data as young. Each point is one donor × cell-state cluster × timepoint pseudobulk from the reprogramming dataset, with at least 20 cells (n = 22). Distance is the nearest-neighbor Euclidean distance to the GTEx training donors in a 50-component PCA space.

state cluster and timepoint, with at least 20 cells), Spearman ρ between nearest-neighbor distance to the GTEx training donors and the frozen score is −0.625 (95% CI −0.834 to −0.320; Figure 5B), and the same sign holds within each donor separately (−0.78 and −0.81). Cells far outside the training data are scored as young. Because distance and reprogramming state move together here, this correlation cannot by itself separate an extrapolation artifact from genuine change, but it does show that a low score cannot be taken at face value.

Being better at age did not protect the ruler. Like MD, it reads movement away from the aged state as rejuvenation, because a single number has no way to register where the cells end up.

DISCUSSION

In the dataset where the distinction can be tested, what is scored as rejuvenation is well explained by departure from the aged state and is not accompanied by a measurable approach to a young cell of the same type. The MD score of the aged donor's partially reprogrammed fibroblasts falls by 7.7 times the difference between that donor and a 22-year-old donor in the same experiment, while in MD's own genes those cells end 2.2 times farther from the young donor's cells than they started. The same holds in the published age gene lists, in all 23,485 genes, in 400 random gene sets, against a GTEx-defined young group, and for a frozen ruler built to track donor age. Young-cell mixtures show the measure would have seen an approach had one occurred, and reverse tests show the movement is not mutual convergence of two donors.

This is not a claim that partial reprogramming fails to rejuvenate. Lu et al. report functional improvements, including in aged mouse liver, that transcriptome distance cannot adjudicate, and the in vivo results in particular lie outside what these data can test. Nor is it a claim that the confound went unnoticed: Lu et al. track cell state explicitly throughout their trajectory. The claim is narrower. A readout of this form cannot separate approach from departure, and in the data examined here the movement is too long to end closer to a young cell of the same type, whatever its direction.

The mechanism is the same in kind across readouts of this form. A curated identity-gene signature such as MD falls whenever the cell type is abandoned, which is why MD tracks position along the reprogramming trajectory well and donor age weakly. A directional score learned by regression can be confounded with distance from its own training distribution, so cells that leave the manifold can score young; we show this for our own ruler rather than only for other people's. A correlation-based reversal score ignores how far a cell moves, so it registers orientation without destination. Changing the genes does not help: the same cells end farther from the young target in every gene set we tested. The problem is the shape of the statistic, not its gene content.

The field already uses the right form of statistic for half of this problem. Cell identity is routinely judged by how closely an engineered population resembles a target cell type, as CellNet (Cahan et al.) and its successors do for reprogramming and direct conversion, and the finding that converted cells retain programs of their starting population is exactly a statement about where they ended up. Age is still judged by direction: a score that falls. Our proposal is that age be evaluated the way identity already is, against a target the intervention is supposed to reach, with the distance to that target reported rather than the sign of a projection. The two then have the same form and can be reported together, which is what a rejuvenation claim needs: this cell type, and young. Distance to a young control has been used before: Roux et al. report that transiently reprogrammed aged mouse cells end closer to young controls. Those cells were profiled after the reprogramming factors were withdrawn, and their displacement is small relative to the age gap; the cells tested here are assayed during OSKM expression and travel more than twice that gap.

The remedy is cheap to adopt. Define the young target on the same

platform as the intervention. Report the change in distance to it alongside progress along a direction. Include a dose-response positive control, so that a null result can be told apart from an insensitive assay, and report the zero-dose point as the noise floor rather than assuming it is zero. Include a reverse test, so that mutual convergence of two samples is not read as rejuvenation of one. Report an identity check alongside the headline number. Check that every reported interval contains its point estimate, and say so when one does not. None of this requires new data: all of the analyses here were performed on public data.

LIMITATIONS

- The reprogramming analysis rests on two donors in one experiment, so donor-level variance cannot be estimated, and in the same-platform test age is confounded with individual. The aged-to-young donor axis is itself nearly orthogonal to the GTEx donor-age axis.
- The partially reprogrammed population tested is dominated by day-3 cells assayed during OSKM expression.
- Cell-state assignments in the reprogramming data are cluster-level signature labels, not lineage tracing. Our labels come from a reimplementation of the published pipeline, run on each donor separately; re-clustering both donors together and using the authors' published labels left every conclusion unchanged, but neither alternative is lineage tracing either.
- The GTEx-anchored positive control ends at the young centroid itself and runs entirely within bulk data, so it shows that a full approach is visible, not that a partial approach across the bulk-to-single-cell gap would be.
- In counts per million without the log, ten genes carry 77% of the squared final distance in the same-platform test, so its agreement with the main analysis rests on few genes. In that coordinate system two transcription-factor perturbations in the unreported CRISPRa screen were labeled as approaching the young centroid, against none in the main analysis. We report them and do not treat them as candidates (Supplementary Note 3).
- The gene-space comparison tests whether the result depends on which genes are read, not whether some other signature could show an approach. The random sets are matched for size, expression level and detection, and none of the 400 showed one, but they are not an exhaustive search over gene sets.
- In the MD genes the published score is a per-cell module score and the geometry a Euclidean distance; the equal-weight direction used to explain why the score falls approximates the published score rather than reproducing it.
- Some distance, cosine and sideways-component intervals in the all-gene and age-gene tests do not contain their point estimates and are flagged invalid; claims that depend on them rest on point estimates. The zero-mixture progress offset in all genes (+0.115) is reported as a noise floor, but its direction was not explained (Supplementary Note 2).
- In the single-cell external fibroblast cohort, an integration of published studies, scores are one value per person (65 donors), and some recorded ages are placeholders. The ruler transfers, but its advantage over MD in that cohort is not distinguishable from zero.
- GTEx donor age is public only in decade bins, so the ruler was trained and evaluated against bin midpoints rather than exact ages.
- Not every number reported here is a gated, pre-specified result: readings were withdrawn and superseded during the project (the superseded text is retained in the archive), and descriptive summaries, namely the per-state scores, the ratios of final to starting distance and the extrapolation correlation, carry no decision rule.
- Transcriptome distance is a molecular proxy. It cannot adjudicate functional rejuvenation, in vivo results, or non-transcriptional mechanisms.
- In the same-platform test the displacement exceeds twice the donor gap, so no approach could have been registered whatever its direction; that test shows overshoot rather than a failed search for approach. The GTEx-anchored test, where an approach was attainable, gives a near-orthogonal result instead.

METHODS

Pre-registration and decision rules

Each gated analysis was specified in writing before any number was computed, including its statistics, nulls, positive controls, decision thresholds and the exact wording of each possible conclusion. A flag file was written before execution and the pre-registration text is reproduced verbatim in the results file for each analysis. Only the pre-registered reading whose conditions fired is reported as the conclusion of that analysis. Where a later analysis superseded an earlier reading, the earlier text was retained and marked withdrawn or superseded rather than edited. Nine readings across the project are marked this way, three of which concern analyses reported here; all are listed in the repository. These are self-administered protocols recorded in the project repository, not deposits with an external registry. Diagnostics and descriptive summaries are not treated as gated results. A published transcription-factor screen was analyzed in the same project and pre-registered in the same way, with its deviation recorded alongside it; it does not bear on the question tested here and is not reported in this paper. Their pre-registrations, code and results, including a transfer that failed its gate, are in the archive.

The ruler

The ruler is a ridge regression of donor chronological age on z-scored gene expression, fit through the n × n Gram matrix, and frozen with its coefficients and its training mean and standard deviation before any evaluation. The regularization parameter was selected over the grid 0.01, 0.03, 0.1, 0.3, 1, 3, 10, 30, 100, 300, 1000, 3000, 10000 by minimizing a leave-one-out residual sum of squares computed from the hat-matrix diagonal, not by grouped cross-validation. The frozen fibroblast ruler stores alpha = 0.01, the smallest value on the grid, so a smaller optimum cannot be excluded. The frozen fibroblast vector is the ridge coefficient vector, unit-normalized and sign-aligned so that the score increases with age. Transfers within GTEx use the full prediction (training mean age plus the ridge coefficients applied to the z-scored profile), which is in years. The external cohorts and all reprogramming analyses use the frozen unit-length score, which orders samples identically but has no intercept or scale and is not in years. Within-site fibroblast cross-validation, grouped by sequencing batch with up to five folds, reports ρ but does not choose alpha. The ruler was never refit on any evaluation cohort.

Training data. GTEx v10 cultured fibroblasts, 652 donors including the seven donors at the small collection site D1, 23,485 genes after filtering. Genes were retained at a count of at least 6 in at least 20% of samples. Counts were TMM-normalized (edgeR-style, log-ratio trim 0.30, absolute-expression trim 0.05, factors scaled to geometric mean 1), converted to log2 counts per million with prior count 2, and z-scored with the training mean and standard deviation; standard deviations below 1e-12 were set to 1. Site D1 is included in the freeze and excluded from the B1 to C1 transfer and from within-site cross-validation, which is stated rather than silently applied.

Donor age. GTEx releases donor age publicly only in decade bins, so every GTEx statistic, including the site transfers and the young and old anchors, uses bin midpoints or bin membership rather than exact ages. The external cohorts provide ages in years, except that some constituent studies of the fibroblast atlas report an age range or a single rounded age for several donors; those recorded values were used as given.

Analysis unit in the fibroblast atlas. The atlas's donor field holds one identifier per sequencing sample in several constituent studies. Each identifier was traced to its source study and the original author's donor identifier, and pseudobulk scores were averaged within each person before any correlation, permutation or bootstrap, giving 65 donors (64 after dropping pseudobulks with fewer than 30 cells, which gave the same conclusion). Seeds and resampling settings were unchanged. This grouping was not pre-registered. The pre-registered analysis treated each of the 93 pseudobulks as a unit and gave $\rho = +0.455$; we report the per-person analysis because repeated samples from one person are not independent, and it gives the lower value. The ruler-minus-MD difference computed on the 93 pseudobulks is superseded by the per-person value in Table 2.

Applying a frozen ruler to a new dataset

All datasets, including the external cohorts and the reprogramming data, are processed identically to the training set: counts summed per group, TMM-normalized, converted to log2 CPM with prior count 2 across that analysis's own pseudobulk panel, then z-scored with the frozen GTEx training mean and standard deviation. Genes absent from the new dataset are set to zero after the frozen z-score, as are genes present but all-zero in that panel. Of the 23,485 ruler genes, 21,351 are present in GSE226189, 17,047 in the fibroblast atlas and 20,452 in GSE297234. Because TMM is estimated within each analysis panel, a statistic computed on that panel is coupled to the panel's membership, which is why bootstrap resamples re-estimate it.

Displacement statistics

For a displacement d from an origin pseudobulk toward a target direction v we report three quantities. Cosine alignment is $d \cdot v / (\|d\| \, \|v\|)$. Progress along the axis is $d \cdot v / \|v\|^2$, which equals 0 at the origin and 1 at the target. The change in distance is $\|z_S - z_target\| - \|z_origin - z_target\|$, so a positive value means the state ended farther from the target. The sideways component is $\|d - progress \cdot v\| / \|v\|$, in units of the axis length. In the same-platform test, where the axis ends at the target, final distance divided by starting distance equals $\sqrt{((1 - progress)^2 + sideways^2)}$, so a state ends closer only if it falls inside the unit circle drawn in Figures 1B and 2.

The two cosines in this paper are not interchangeable and are never combined. In the GTEx-anchored test the target direction is the unit vector from the old centroid to the young centroid, and the reported change in distance is to the young centroid. In the same-platform test the axis runs from one donor's day-0 cells to the other donor's day-0 cells, and progress is the projection defined above. This distinction matters numerically: the change in distance at pluripotency is +184 on the GTEx axis and +382 within the experiment, and those are two different measurements of two different things.

In the GTEx-anchored test, a state counts as approaching young only if its alignment is positive and beats both nulls and its distance to the young centroid decreases; in the same-platform test, only if its distance to the young target decreases. Alignment or progress alone is reported as departure from old, not as approach.

Cell-state labels

Cells were assigned to five states (fibroblast, partially reprogrammed, early pluripotent, pluripotent and non-reprogrammed) by Louvain clustering of each donor's cells separately, with a Python reimplementation of the published pipeline (log-normalization with regression on the mitochondrial fraction, 3,000 variable genes, 50 principal components, resolution 0.8; the published pipeline uses sctransform in Seurat, which was not available), and labeling each cluster with the published state signatures. States are pooled across timepoints within each donor. Two alternative labelings were used as checks. Re-clustering both donors together, as Lu et al. did, changed the label of 7.8% of the aged donor's cells and 8.3% of the young donor's. The authors' published per-cell states (16 sub-states in the combined object deposited with GSE297234, grouped into the five states by name) agree with ours for 80% of the aged donor's cells and 72% of the young donor's; cells absent from the published object were excluded. Where day-0 fibroblast membership changed, the day-0 half-split was redrawn with the same seed. With the joint labels, the change in distance to the young target was +133.3 in all genes and +17.5 in the MD genes (original labels +130.9 and +17.1); with the published labels, +135.8 and +17.8. In every case the mixture control passed and the interval on the forward-minus-reverse asymmetry excluded zero, and the MD drop was 7.8 and 8.0 times the donor gap (original 7.7). The GTEx-anchored change in distance to the young centroid was +9.4 with the joint labels and +15.8 with the published labels (original +9.4).

Signature scores and gene spaces

Signature scores. MD, the age-up and age-down lists and the cell-state signatures are the published gene lists, scored with a Python reimplementation of the Seurat AddModuleScore algorithm on log-normalized counts (24 expression bins, 100 control genes per bin, scale factor 10,000). For the per-state MD scores reported above and in Figure 1A, the two donors were placed on a single gene axis and one set of bins and control genes was drawn from the pooled cells of both donors, so that the two donors' scores are on the same scale; the source pipeline scores each donor separately, which widens the ratio of the reprogramming drop to the donor gap from 7.7 to 13.0 and does not otherwise change the result.

Gene spaces. A gene space is a subset of the reference panel's 23,485 genes. Every space uses the same TMM-normalized, frozen z-scored matrix, with the columns selected afterwards, so normalization is never re-estimated on a subset. Published symbols

are matched to the panel's symbols exactly after trimming and upper-casing, with no alias resolution: 205 of 205 MD genes, 1,312 of 1,533 age-up genes and 1,777 of 2,007 age-down genes map, the two age lists do not overlap, and 43 genes are shared between MD and the age lists. Within each space the origin, the target, the axis, the mixture control and the reverse test are rebuilt from that space's own columns rather than projected from the full space. Cosines and distances are not comparable between spaces of different dimension, so every cross-space statement uses progress, the perpendicular component in axis lengths, the ratio of final to starting distance, or the random-set baseline. Random sets are drawn without replacement within strata defined by 24 equal-count bins of the frozen GTEx training mean crossed with whether the gene has any reads in the panel, so that a random set of a given size has the same expression profile and the same number of undetected genes as the real set it is compared with.

Nulls, intervals and multiple testing

All nulls use 200 draws and an empirical p-value of (exceedances + 1) / 201. Three constructions appear, and they are not interchangeable. A donor-label permutation shuffles donor age within the training bank, holding the expression matrix fixed, and is the null for every transfer statistic. A gene-permutation null shuffles the entries of the frozen target direction across genes, and tests whether a cosine is larger than for a direction with the same coefficient distribution. An anchor null rebuilds the young-minus-old direction from equal-size random splits of all GTEx fibroblast donors, ignoring age, and tests whether an alignment is specifically with age rather than with any donor contrast.

Confidence intervals are 200-draw percentile bootstraps at the 2.5th and 97.5th percentiles. The resampling unit is the donor for transfer statistics and the cell for displacement statistics. Paired comparisons recompute both quantities on the same resampled donors, so that the interval is on the difference rather than on two separate intervals. Any interval that does not contain its own point estimate is flagged invalid and is not used as a gate; this happens for distance, cosine and sideways-component intervals in the high-dimensional displacement tests and is reported at each occurrence.

Seeds are fixed: 20260914 for splits, permutations and null directions, 20260918 for bootstraps, with an independent generator per cell.

Positive controls

No test is interpreted until a control has recovered a known effect, and the controls are rebuilt inside each gene space rather than carried over from the full space. Two kinds are used. In the same-platform test, synthetic mixtures combine known fractions of young day-0 cells into aged day-0 cells at fixed total cell count; the control passes if progress increases monotonically with the mixing fraction, the interval on progress at the highest fraction excludes zero, and the change in distance at that fraction is negative. The zero-mixture point is reported as the noise floor and is not subtracted. In the GTEx-anchored test, five folds displace held-out old donors toward the training fold's young centroid; the bar was every fold cosine above 0.5, and the observed minimum was 0.615 in all genes, 0.733 in the MD genes and 0.681 in the age genes.

Datasets

Table 3. Datasets used in this paper. All are public. Accession-level detail and per-cohort sample tables are in the repository.

Dataset	Identifier	Use here
GTEx v10 cultured fibroblasts	GTEx Portal release v10 (dbGaP phs000424)	Ruler training (652 donors); young and old anchors (113 and 231 donors)
Consensus dermal fibroblast atlas	CELLxGENE a19d1667-a7b5-4556-9e5f-f9bfa690c0f1	External fibroblast transfer, 65 donors, 10x assays only
Human skin fibroblast bulk RNA-seq	GSE226189	External fibroblast transfer, 82 donors
OSKM reprogramming time course	GSE297234	Reprogramming tests (two donors, days 0, 3, 7, 10)
Gene sets	MSigDB Hallmark EMT; Lu et al. supplementary tables	MD signature; age-up and age-down lists; cell-state signatures

Software

Analyses were implemented in Python 3.12.7 with numpy 2.5.3, scipy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, anndata 0.13.3.post0 and h5py 3.16.0, and figures were drawn with matplotlib 3.11.2. Versions were recorded from the analysis environment at the time of writing; the full listing is in the repository. Entry-point scripts, one per analysis, are listed in the repository README and reproduce every table in this paper from the frozen instruments and the public inputs. Seeds are given above. Seurat v5 and sctransform were not available in this environment; the AddModuleScore re-implementation (Signature scores, above) is a deviation from the published MD pipeline rather than a substitution.

DATA AND CODE AVAILABILITY

All input data are public and are listed in Table 3. Analysis code, the frozen instruments, the pre-registration documents with their timestamp flags, and every result table used in this paper are available at https://github.com/k-dlrgus/rejuvenation-readouts and archived at https://doi.org/10.5281/zenodo.23092318. The frozen fibroblast ruler is a single file containing the unit weight vector, the un-normalized coefficients and the training mean and standard deviation over 23,485 genes; the young and old GTEx anchors are provided as a separate file. The repository README lists the entry-point script that reproduces each table and figure. Raw input data are not redistributed and must be obtained from the sources in Table 3.

AUTHOR CONTRIBUTIONS

R.K. is the sole author. He conceived the study, specified the analyses and their pre-registered decision rules, directed their execution, and is responsible for the manuscript.

USE OF AI TOOLS

Analysis code was written by AI coding agents under the author's

written pre-registration protocols, and the manuscript was drafted with AI assistance. The author reviewed all analyses and takes full responsibility.

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SUPPLEMENTARY INFORMATION

Supplementary Note 1. Young defined by GTEx donors

Anchors were the mean expression of the GTEx reference donors aged 20 to 39 (n = 113) and 60 to 79 (n = 231), with the 40 to 59 bins excluded. In all genes, the aged donor's partially reprogrammed displacement is almost orthogonal to the young-minus-old axis (cosine +0.077). That small alignment survives a gene-permutation null (p = 0.005) but not a null in which GTEx donors are split at random ignoring age (p = 0.39), so it is not specifically an age alignment. The cells end farther from the young centroid (+9.4) and farther from the old centroid (+10.9): they recede from both. A five-fold positive control, in which held-out old GTEx donors are displaced toward the young centroid, gave fold cosines of 0.615 to 0.874, above the bar of 0.5, so the statistic registers a genuine approach when one occurs (Supplementary Figure S1D); the control also passes in the MD genes (0.733 to 0.895) and the age genes (0.681 to 0.895).

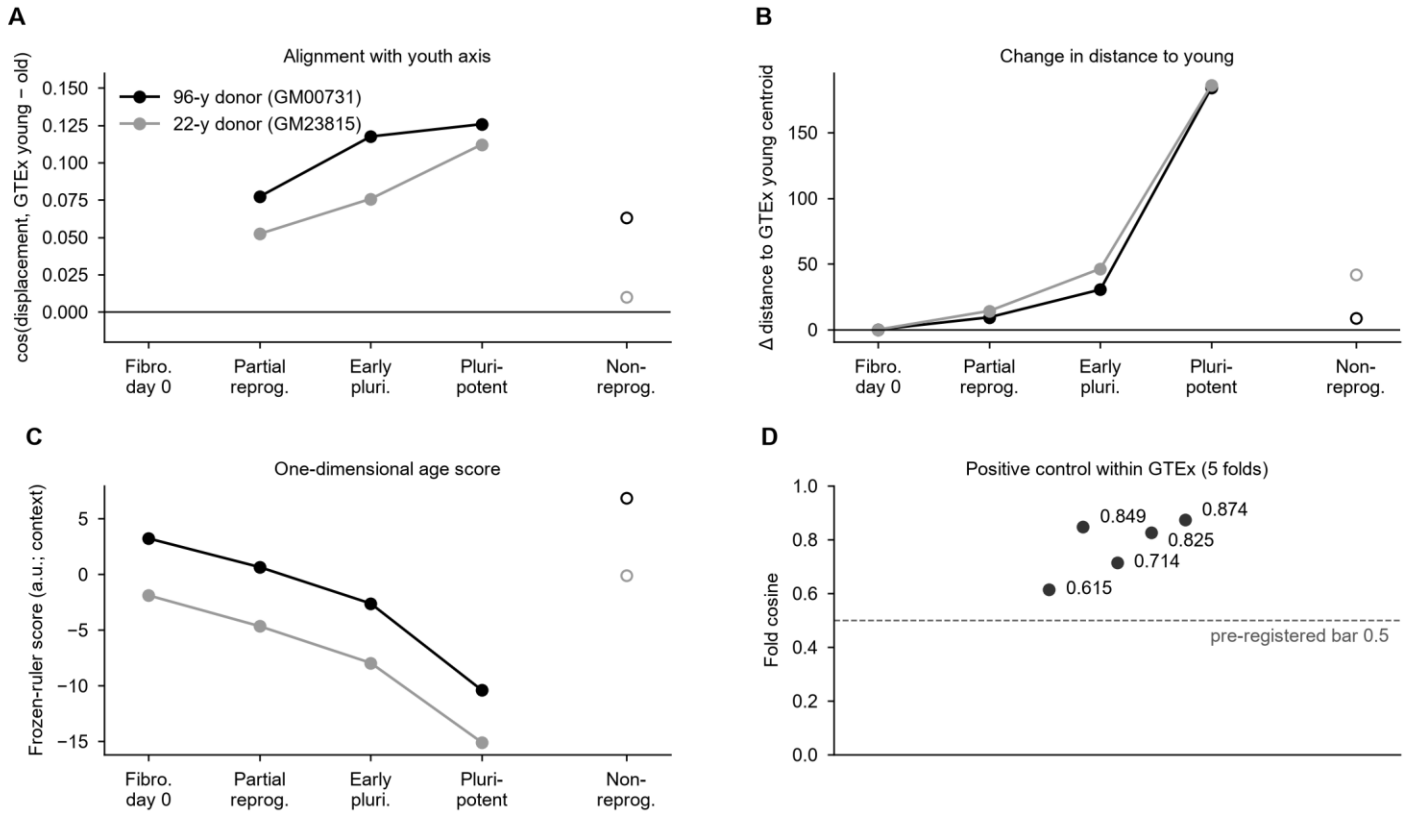
That control has a limit worth stating. It runs entirely within GTEx bulk data and ends at the young centroid itself, whereas the reprogramming origin is a single-cell pseudobulk that already sits 467 units from the young centroid while the young and old centroids are only 32 apart (42.0 and 3.3 in the MD genes; 164.0 and 12.3 in the age genes). The GTEx-anchored distances are therefore dominated by the platform gap, and the control does not show that a partial approach would be visible across that gap. This is why the main test uses a young target measured in the same experiment.

Along the full trajectory, cosine alignment with the youth axis rises from partially reprogrammed to early pluripotent to pluripotent cells (+0.077, +0.117, +0.126), while the change in distance to young fibroblasts rises far faster (+9.4, +30.5, +184.0) and the ruler score falls (+0.65, −2.62, −10.40). The farther the cells travel from the aged fibroblast state, the younger they look on a directional score and the less they resemble a young fibroblast. The distance traveled is not monotonic across these states (271.7, 268.6, 546.4 units), so cosine order and displacement order are not the same ordering.

Supplementary Note 2. The same-platform test in all genes

In all 23,485 genes, the young-cell mixtures at 0%, 10%, 25% and 50% produced monotonically increasing progress (0.115, 0.182, 0.283, 0.441) and monotonically decreasing distance (+0.07, −7.87, −18.91, −34.44), so the statistic could have detected an approach had one occurred. Two caveats attach to that control. The zero-mixture point does not sit at zero progress (+0.115); the anchors are built from one random half of the day-0 cells and the mixtures from the other, so some offset is expected, but why it leans toward the young target was not resolved by our diagnostics, and it is reported as the statistic's noise floor rather than subtracted. In the MD genes the same offset is 0.010. And in all genes the bootstrap intervals on the distance values do not contain their point estimates and are flagged invalid, so there the monotonicity claim rests on the point estimates alone; the intervals on progress, used as the gates, are valid.

With the roles of the donors reversed, the young donor's partially reprogrammed cells do not move toward the aged donor at all in all genes (progress −0.057, 95% CI −0.063 to −0.054), giving a clear asymmetry (+0.732, CI +0.724 to +0.740). The forward movement is therefore not symmetric mutual convergence. But it is also not obviously age: the aged-to-young donor axis is itself nearly orthogonal to the GTEx donor-age axis (cosine 0.039), so with one donor at each age we cannot separate age from everything else that differs between two cell lines, including proliferative state and culture history. The test can, and does, say that the cells did not approach the young line.



Supplementary Figure S1. Reprogramming states against a GTEx-defined youth axis, all genes. (A) Cosine between each state's displacement from day-0 fibroblasts and the GTEx young-minus-old direction. (B) Change in Euclidean distance to the GTEx young centroid; positive is farther. (C) The frozen ruler score of each state, shown as context. (D) Positive control: the 652 GTEx fibroblast donors were split into five folds; in each fold, held-out old donors were displaced toward that fold's young centroid. Open symbols in A to C are non-reprogrammed cells (OSKM-exposed cells that remained off the reprogramming path), not a second day-0 sample. Donors are never pooled; the 22-year-old donor is a contrast, not a replicate.

Supplementary Note 3. The same-platform test in three coordinate systems

The same-platform test was repeated in three coordinate systems that differ only in the last step of the transform. In space A, coordinates are TMM-normalized counts per million, with no log and no z-score, so abundant genes dominate. In space B they are log2 CPM with prior count 2, the matrix the main analysis z-scores, so distances are in fold-change units. Space C is the main analysis: log2 CPM z-scored with the frozen GTEx training mean and standard deviation. Everything upstream of that step was shared: the pseudobulk panel, the summed counts, the TMM factors, the day-0 half-split, the young-cell mixtures, and the bootstrap and permutation draws. The frozen ruler was not refit in any space; only the distance test moves. Genes absent from the panel contribute nothing to any within-panel difference in any space. The analysis was pre-registered before any number in space A or B existed, with space C fixed as the primary, the qualification rule below, and a decision table for each outcome. Before any space A or B number was computed, space C was rebuilt with the new code and reproduced every stored statistic of the main analysis to within rounding.

Qualification. A space counts only if, in that space, the young-cell mixture control detects known young fractions. The criteria are those used throughout the paper. Progress must rise monotonically across 0, 10, 25 and 50% young cells. The interval on progress at 50% must exclude zero, and must also contain its own point estimate. The change in distance at 50% must be negative. A space that failed would still have been run to the end and reported, with no conclusion drawn from it. All three spaces qualified. The zero-mixture noise floor in progress was 0.039 in A (95% CI −0.009 to 0.085), 0.106 in B and 0.115 in C. In A the zero-mixture change in distance is −498 CPM units, 9% of the change at 50% young cells (−5,402), and is reported as a floor, not subtracted. Qualification was written to a timestamped flag before any partially reprogrammed vector was built in space A or B.

Supplementary Table S1. The GTEx-anchored test in three gene spaces, aged donor (GM00731). Change in Euclidean distance from the donor's day-0 fibroblasts to the GTEx young (20 to 39 years) and old (60 to 79 years) centroids, restricted to each gene space; positive is farther. The last column is the lowest fold cosine of the five-fold positive control in that space (pass bar 0.5). In every space the day-0 cells start far from both centroids relative to the young-to-old gap (all genes 467.5 against 32.3; MD genes 42.0 against 3.3; age genes 164.0 against 12.3), so this test shows departure but could not show a small approach.

Gene space	Cell state (aged donor)	Change in distance to GTEx young	Change in distance to GTEx old	Five-fold control, lowest fold cosine
All genes (23,485)	Partially reprogrammed	+9.4	+10.9	0.615
	Early pluripotent	+30.5	+32.7	
	Pluripotent	+184.0	+188.3	
	Non-reprogrammed	+8.9	+9.8	
MD genes (205)	Partially reprogrammed	+6.5	+7.0	0.733
	Early pluripotent	+7.4	+8.0	
	Pluripotent	+28.2	+29.2	
	Non-reprogrammed	−1.4	−1.3	
Age genes (3,089)	Partially reprogrammed	−5.4	−5.9	0.681
	Early pluripotent	+1.7	+2.1	
	Pluripotent	+44.7	+45.9	
	Non-reprogrammed	−1.0	−0.7	

Supplementary Table S2. The same-platform test in three coordinate systems, aged donor (GM00731), partially reprogrammed cells pooled across timepoints (n = 5,832). Mixture: progress at 50% young cells, with its interval, and the change in distance at 50%. Change in distance is in the units of each space and is not comparable across rows. Final/start is the distance to the young donor's day-0 cells after, divided by before. Top-10 share is the fraction of the squared final distance carried by the ten genes with the largest squared coordinates.

Space	Coordinates	Mixture, progress at 50% (95% CI)	Mixture, change in distance at 50%	Progress (95% CI)	Change in distance	Final / start	Reverse progress (95% CI)	Asymmetry (95% CI)	Top-10 share
A	CPM	0.417 (0.371 to 0.459)	−5,402	2.045 (2.026 to 2.052)	+17,021	2.05	−1.208 (−1.213 to −1.195)	+3.253 (+3.224 to +3.261)	77.3%
B	log2 CPM	0.461 (0.446 to 0.466)	−30.7	0.581 (0.574 to 0.583)	+99.4	1.88	+0.074 (+0.070 to +0.078)	+0.507 (+0.498 to +0.511)	3.6%
C	log2 CPM, z-scored (main analysis)	0.441 (0.424 to 0.447)	−34.4	0.674 (0.667 to 0.680)	+130.9	2.00	−0.057 (−0.063 to −0.054)	+0.732 (+0.724 to +0.740)	1.8%

In every space the cells travel a long way along the donor axis and still end farther from the young donor than they started: no space flips the sign. Every interval used as a gate contains its point estimate. The intervals on the change in distance of the mixtures, at every mixing fraction, do not contain their point estimates in spaces B and C, and neither do those of the pluripotent and non-reprogrammed context states. None of these is a gate, and this is the invalid-interval pattern already reported for the main analysis (Supplementary Note 2). In space A all intervals contain their point estimates. The intervals on the change in distance of the partially reprogrammed cells contain their point estimates in all three spaces. In space B the young donor's partially reprogrammed cells move slightly toward the aged donor (reverse progress +0.074), but much less than the aged donor's cells move toward the young one, so the asymmetry stays positive. In the main analysis the reverse movement was absent.

How many genes carry the distance. Euclidean distance adds over genes only in its square, so we report the share of the squared distance carried by the ten largest squared coordinates. In space C the starting and final distances are spread across the genome: the top ten genes carry 2.2% and 1.8%, and the leading genes of the final distance are reprogramming genes (POU5F1, SOX2, KLF4). Space B is similar (3.4% and 3.6%). In space A the top ten genes carry 72.7% of the starting distance and 77.3% of the final distance: MALAT1, KLF4, VIM, LGALS1, POU5F1, S100A6, FTL, TMSB10, COL1A2 and TMSB4X. Without the log, the same-platform result is a statement about a handful of highly expressed transcripts. It agrees in sign with the other two spaces, but it does not carry the same weight. The pre-registration predicted this risk for space A in this test, where the cell counts are in the thousands and sequencing depth is not the limiting factor.

The transcription-factor screen. The pre-registered check also covered the published CRISPRa screen that the Methods describe as analyzed but not reported here. The same rule applied: the 1,836 perturbations, the shuffled-vector floors, the gates, and the GTEx old-to-young comparison, with space C as the primary. The protocol required any change to be disclosed, so we report it here. In spaces B and C no perturbation is labeled as approaching the young centroid. In space A two are: ZNF441 and ZNF669, with changes in distance of −4.5 and −69.5 CPM units against a GTEx old-to-young distance of 8,225 (q = 0.005 for both). Several diagnostics show what space A does to this test:

- Ten genes carry 94% of the squared GTEx old-to-young distance in space A: FN1, SERPINE1, COL1A2, TGFBI, FTL, MT-ND2, COL1A1, MT2A, COL3A1 and ANXA2. They also carry a median 90% of each perturbation's squared final distance (5th to 95th percentile, 85% to 92%). In space C the same shares are 0.6% and 4%.
- Every one of the 1,836 endpoints has negative coordinates, 3.5 million in total. This happens because space A adds a CPM difference measured in Hs27 cells to a CPM origin built from GTEx donors. The coordinates were kept as they are, not clipped.
- The ranking in space A agrees poorly with the main analysis (Spearman 0.24 across all perturbations; 1 of the top 20 shared). Space B agrees closely (0.99; 14 of 20).
- In space C both perturbations move away from the young centroid (+106.6 and +113.8, q = 1.0), and in space B both end farther from it (+32.9 and +35.1). They have 84 and 82 cells, against a median of 106.
- ZNF441's change in distance does not clear the non-targeting split floor in space A (−14.2), which is not a gate for the label. ZNF669's does.

Space C remains the screen's pre-registered primary, and in it no perturbation approaches. In space C, for the GTEx old-to-young comparison, with the GTEx old centroid (60 to 79 years) as the origin and the GTEx young centroid (20 to 39 years) as the target, every one of the 1,836 perturbations moved more than twice the distance between the two centroids (smallest ratio 2.10; per-perturbation values in results/south6/ in the repository), so from that origin no approach to that target could have been registered whatever its direction. This holds for that origin and target, which sit only 32 units apart, and is not a statement about the screen in general; it is the same limitation the main text discloses for the same-platform test. What follows is a post-hoc reading, not a pre-registered conclusion, and it carries no decision rule: the protocol required only that the space A labels be reported and that space C remain the primary. On the diagnostics above, which show that the two space A labels rest on ten genes and on negative coordinates, we do not treat them as candidates.

The pre-registration, the space C reproduction table, the per-space mixture controls, the full per-perturbation table in all three spaces and the decision record are in the repository (results/coordspace/).